

CENSORED REGRESSION QUANTILES*

James L. POWELL

*Massachusetts Institute of Technology, Cambridge, MA 02139, USA
University of Wisconsin, Madison, WI 53706, USA*

The object of this paper is to demonstrate how the LAD estimation method for the censored regression model can be extended to more general quantiles. In this paper, the form of the conditional quantiles for the censored regression models is heuristically derived and discussed. The resulting estimators of the regression coefficients, which include the censored LAD estimator as a special case, are shown to be consistent and asymptotically normally distributed under appropriately 'translated' versions of the corresponding assumptions for the former approach; consistent estimation of the asymptotic covariance matrix when the error terms are i.i.d. is also treated. The paper discussed how several quantile estimators might be combined to improve efficiency when the error terms are identically distributed, and how tests of homoskedasticity and symmetry of the error distribution can be constructed using particular differences of the estimated coefficients.

1. Introduction

Semi-parametric estimation of non-linear models has been the focus of much recent attention in theoretical econometric research. Likelihood-based procedures for non-linear econometric models – specifically, qualitative and limited dependent variable models, general sample selection models, and other models in which the unobserved error terms enter non-linearly in the generation of the observable data – are very sensitive to misspecification of the error distribution. Since economic theory typically yields no restrictions on the parametric form of these distribution functions, semi-parametric estimation of regression coefficients under weak conditions on the distribution of the error terms has an obvious practical appeal. While the literature has (justifiably) focused upon consistency of proposed estimation methods, the related question of best attainable efficiency when the error distribution has unknown parametric form has also been addressed.

Several distinct approaches to semi-parametric estimation of non-linear models have been taken in the literature. The approach to be taken in this paper uses conditional quantiles, rather than mathematical expectations, to characterize the distribution of the observable data in terms of the parameters and error distribution. Quantile-based estimators have long been known in the

*This research was supported by National Science Foundation Grants SES79-12965 and SES79-13976 at Stanford University and Grant SES83-09292 at M.I.T. I would like to thank Takeshi Amemiya, Theodore W. Anderson, Timothy Bresnahan, Colin Cameron, Thomas MaCurdy, and Daniel McFadden for their helpful comments.

statistical literature (as ‘*L*-estimators’, i.e., linear combinations of order statistics) for their relative efficiency for heavy-tailed error distributions; analysis of (essentially) linear models using quantile estimation dates from the study by Koenker and Bassett (1978), and has since been extended to heteroskedastic linear models [Koenker and Bassett (1982)], time series models [Bloomfield and Steiger (1983)], the non-linear regression model [Obenhofer (1983)], and linear simultaneous equations models [Amemiya (1982), Powell (1983)]. Nevertheless, an important characteristic of population quantiles – namely, their equivariance with respect to monotonic transformations of the associated random variable – has only recently been exploited in the analysis of non-linear econometric models. The ‘maximum score’ estimator proposed by Manski (1975) for the binary choice model used the conditional median of the response variable to represent the ‘typical response’ in terms of the regressors and unknown parameters, and this approach has recently been extended [Manski (1984)] to estimators based on more general quantiles. Similarly, estimation of regression parameters using a least absolute deviations (LAD) criterion has recently been proposed by Powell (1984) for the linear regression model when the dependent variable is censored; as in the original maximum score paper, this estimation method ‘centered’ the conditional distribution of the observed dependent variable at its median.

The object of the present paper is to demonstrate how the LAD estimation method for the censored regression model can be extended to more general quantiles. In the following section, the form of the conditional quantiles for the censored regression models is heuristically derived and discussed. In section 3, the resulting estimators of the regression coefficients, which include the censored LAD estimator as a special case, are shown to be consistent and asymptotically normally distributed under appropriately ‘translated’ versions of the corresponding assumptions for the former approach; consistent estimation of the asymptotic covariance matrix when the error terms are i.i.d. is also discussed. The final section discusses how several quantile estimators might be combined to improve efficiency when the error terms are identically distributed, and how tests of homoskedasticity and symmetry of the error distribution can be constructed using particular differences of the estimated coefficients.

2. Definitions and motivation

The censored regression model, or censored ‘Tobit’ model [after Tobin (1958)], can be written in the form

$$y_i = \max\{0, x_i'\beta_0 + u_i\}, \quad i = 1, \dots, n, \quad (2.1)$$

where the dependent variable y_i and the regression vector x_i are observed for each i , while the parameter vector β_0 and error term u_i are not observed. For

this model, the censored LAD estimator $\hat{\beta}_n$ was defined in Powell (1984) to be that value of β minimizing

$$S_n(\beta) = \frac{1}{n} \sum_{i=1}^n |y_i - \max\{0, x_i'\beta_0\}|, \quad (2.2)$$

over all β in some parameter space B . This definition was based upon the simple relationship between the median of the censored dependent variable and the regressors and parameter vector; hence this estimation method is just median (LAD) regression for y_i using the non-linear regression function $\max\{0, x_i'\beta_0\}$. But neither the median nor any other measure of location will completely characterize the conditional distribution of y_i given the regressors x_i (except in special cases), so other statistics can be expected to provide additional information about the unknown coefficients β_0 .

Regarding the median as the 50th percentile of the distribution of y_i , it is natural in this context to consider the relationship of other percentiles of y_i to the underlying regression function $x_i'\beta_0$. As argued below, the median is not the only percentile to bear a simple functional relationship to the unknown parameters in the censored regression model, and calculation of other percentiles may reveal a great deal about β_0 . Indeed, if the dependent variable is heavily censored – so that $y_i = 0$ for a large proportion of the observations – the censored LAD estimator $\hat{\beta}_n$ may be very imprecise, since the median of y_i would be uninformative about β_0 for much of the sample. In such a situation, it may be preferable to ‘center’ the distribution of y_i at a higher quantile than the median, because a higher quantile would be more often positive, and thus more often informative about β_0 .

Estimation of sample percentiles for i.i.d. random variables has recently been extended to the linear model by Koenker and Bassett (1978, 1982); their proposed estimators of regression coefficients include LAD regression as a special case. Koenker and Bassett have shown that, apart from an intercept term, each ‘regression quantile’ estimator is consistent for the unknown coefficients, and estimators for different quantiles are jointly asymptotically normal. Here, the theory of regression quantiles is extended to the censored regression model in the same way that LAD estimation was previously generalized to account for censored samples.

The first step in the definition of ‘censored regression quantile’ estimators is the determination of the functional form of the quantiles for the dependent variable y_i . Let $F_Y^{-1}(\theta|x_i, \beta_0)$ denote the θ th quantile of y_i for $\theta \in (0, 1)$; for example, $F_Y^{-1}(\frac{3}{4}|x_i, \beta_0)$ would be the upper quartile of y_i given x_i and β_0 , while $F_Y^{-1}(\frac{1}{2}|x_i, \beta_0)$ is the conditional median (50th percentile). Suppose the distribution function $F(\lambda)$ of the error terms $\{u_i\}$ is continuously differentiable with positive density at $F^{-1}(\theta)$, the θ th quantile of the i.i.d. sequence u_i ; then the equivariance of population percentiles with respect to monotonic

transformations implies that the θ th conditional quantile of y_i is just the value of the right-hand side of (2.1) when u_i is replaced by its θ th quantile, i.e.,

$$F_Y^{-1}(\theta|x_i, \beta_0) = \max\{0, x_i'\beta_0 + F^{-1}(\theta)\}. \quad (2.3)$$

The median of y_i is thus the special case of (2.3) derived previously, provided the median of u_i , $F^{-1}(\frac{1}{2})$, is zero. Furthermore, if the first component of the parameter vector β_0 is an intercept term (so that $x_{i1} \equiv 1$), the θ th quantile of y_i can be rewritten as

$$F_Y^{-1}(\theta|x_i, \beta_0) = \max\{0, x_i'\beta_0(\theta)\}, \quad (2.3')$$

where

$$\beta_0(\theta) \equiv \beta_0 + F^{-1}(\theta) \cdot e^1, \quad e^1 \equiv (1, 0, \dots, 0)'; \quad (2.4)$$

hence each quantile of y_i varies with the non-constant regressors in the same way the median does.

To use data from a censored regression model to obtain an estimator of $\beta_0(\theta)$ for a particular value of θ , a minimand analogous to $S_n(\beta)$ in (2.2) is required. The proper 'loss function' to be minimized in this case is given by Koenker and Bassett (1978), who use the fact that the θ th quantile of a random variable Z minimizes $E[\rho_\theta(Z - b) - \rho_\theta(Z)]$ over b , where

$$\rho_\theta(\lambda) \equiv [\theta - 1(\lambda < 0)] \cdot \lambda, \quad (2.5)$$

for $1(A)$ denoting the indicator function of the event A (i.e., it takes the value one if A is true, and is zero otherwise). Thus the censored regression quantile estimator $\hat{\beta}_n(\theta)$ of $\beta_0(\theta)$ is defined as that value of β minimizing

$$Q_n(\beta; \theta) \equiv \frac{1}{n} \sum_{i=1}^n \rho_\theta(y_i - \max\{0, x_i'\beta\}), \quad (2.6)$$

over all β in some parameter space $B(\theta)$. Again, the censored LAD estimator $\hat{\beta}_n$ is easily seen to be a special case; $Q_n(\beta; \frac{1}{2}) = S_n(\beta)/2$, so $\hat{\beta}_n \equiv \hat{\beta}_n(\frac{1}{2})$, as defined above.

The conditions under which minimization of $Q_n(\beta; \theta)$ of (2.6) will yield a consistent estimator of β_0 will thus be analogous to, but more general than, the corresponding conditions for censored LAD estimation. Of course, the intercept term in β_0 is not identified (and thus cannot be consistently estimated) without some further restriction on the distribution of the $\{u_i\}$, e.g., normalizing the θ th quantile of u_i to be zero for some fixed θ (for the analysis of the censored LAD estimator, the median of u_i was taken to be

zero). For the slope (i.e., non-intercept) coefficients in β_0 , consistency of the θ th regression quantile estimator will require the θ th quantile of the error distribution to be uniquely defined (more precisely, the error distribution will be assumed to be absolutely continuous with positive density at the θ th quantile). Also, the conditional quantile function of (2.3) above must be informative about the parameter vector, which requires $x'_i\beta_0 + F^{-1}(\theta) > 0$ for a positive fraction of the observations, and the regressors which satisfy this condition must be sufficiently variable to identify β_0 . As discussed above, taking values of θ closer to unity helps ensure that the latter condition is satisfied, but such quantiles of the error distribution can typically be estimated less precisely, so optimal choice of θ will be difficult to determine, as discussed in section 4 below.

3. Large sample behavior of the estimators

Consistency and asymptotic normality of $\hat{\beta}_n(\theta)$ can be demonstrated under appropriately 'translated' versions of the conditions used to obtain the same properties for the censored LAD estimator [given in Powell (1984)]. Thus, to show consistency of $\hat{\beta}_n(\theta)$ for a particular value of θ , the following assumptions will suffice:

Assumption P.1. The parameter vector $\beta_0(\theta)$ [given in (2.3') above] is an element of a compact parameter space $B(\theta)$.

Assumption R.1. The regressors $\{x_i\}$ are independently distributed random vectors with first component $x_{i1} \equiv 1$ (a.s.), $E\|x_i\|^3 < K_0$ for all i and some positive K_0 , and ν_n , the smallest characteristic root of the matrix

$$E\left\{\frac{1}{n} \sum_{i=1}^n 1(x'_i\beta_0(\theta) \geq \epsilon_0) \cdot x_i x'_i\right\},$$

has $\nu_n > \nu_0$ whenever $n > n_0$, for some positive ϵ_0 , ν_0 , and n_0 (possibly depending upon θ).

Assumption E.1. The error terms $\{u_i\}$ are independent and identically distributed random variables with distribution function $F_u(\lambda)$ which is continuously differentiable with positive density $f_u(\lambda)$ at $\lambda = F_u^{-1}(\theta)$.

Theorem 1. For the censored regression model (2.1), if Assumptions P.1, R.1, and E.1 hold for a particular value of θ , the censored regression quantile

estimator $\hat{\beta}_n(\theta)$ of $\beta_0(\theta)$ is strongly consistent; that is, if

$$\min_{\beta \in B(\theta)} Q_n(\beta; \theta) \equiv Q_n(\hat{\beta}_n(\theta); \theta),$$

then $\hat{\beta}_n(\theta) \rightarrow \beta_0$ almost surely.

Proof. The proof of Theorem 1 of Powell (1984), which showed consistency of the censored LAD estimator $\hat{\beta}_n$, can easily be adapted to the present case. By Lemmas 2.2 and 2.3 of White (1980), strong consistency of $\hat{\beta}_n(\theta)$ will hold if

$$\bar{Q}_n(\beta; \theta) \equiv E[Q_n(\beta; \theta) - Q_n(\beta_0(\theta); \theta)] \quad (3.1)$$

is strictly positive for $\|\beta - \beta_0\| > \varepsilon$ for arbitrary $\varepsilon > 0$ and all n sufficiently large. But it can be shown that

$$\begin{aligned} \bar{Q}_n(\beta; \theta) \geq E_x \left[\frac{1}{n} \sum_{i=1}^n 1(x'_i \beta_0(\theta) \geq 0, x'_i \beta \geq 0) \right. \\ \times \int_0^{x'_i \delta} (x'_i \delta - \lambda) \cdot f_u(\lambda - F_u^{-1}(\theta)) d\lambda \\ \left. + \frac{1}{n} \sum_{i=1}^n 1(x'_i \beta_0(\theta) \geq 0, x'_i \beta < 0) \right. \\ \left. \times \int_{-x'_i \beta_0(\theta)}^0 (\lambda + x'_i \beta_0(\theta)) \cdot f_u(\lambda - F_u^{-1}(\theta)) d\lambda \right], \quad (3.2) \end{aligned}$$

where 'E_x' denotes expectation taken with respect to the distribution of the regressors $\{x_i\}$ and $\delta \equiv \beta - \beta_0(\theta)$. [This reduces to inequality (A.9) of Powell (1984) when $\theta = \frac{1}{2}$.] Assumptions R.1 and E.1 and inequality (3.2) can be used to show $\bar{Q}_n(\beta; \theta)$ is uniformly positive for β with $\|\beta - \beta_0(\theta)\| > \varepsilon$ and sufficiently large n , using the same argument as in (A.10) and (A.11) of Powell (1984). ■

To prove asymptotic normality of the censored LAD estimator, stronger conditions on the parameter space, regressors, and error terms were imposed,

and analogous conditions in the more general quantile setting are easily obtained:

Assumption P.2. The parameter vector $\hat{\beta}(\theta)$ is an interior point of the (compact) parameter space $B(\theta)$.

Assumption R.2. In addition to the conditions of Assumption R.1, defining

$$G_i(z, \beta, r) \equiv E_x[1(|x'_i\beta| \leq \|x_i\| \cdot z) \|x_i\|^r],$$

the function G_i is $o(z)$ for z near zero, β near $\beta_0(\theta)$, and $r \leq 2$, uniformly over i ; that is,

$$G_i(z, \beta, r) \leq K_1 \cdot z,$$

if $0 \leq z < \zeta_0$, $\|\beta - \beta_0(\theta)\| < \zeta_0$, and $r \leq 2$, for some K_1 and ζ_0 .

Assumption E.2. In addition to the conditions of Assumption E.1, the density function $f(\lambda)$ of the $\{u_i\}$ is Lipschitz continuous,

$$|f(\lambda_1) - f(\lambda_2)| \leq L_0 \cdot |\lambda_2 - \lambda_1|,$$

for some positive L_0 .

Let $\hat{\xi}_n$ denote a vector of J regression quantile estimators,

$$\hat{\xi}_n \equiv (\hat{\beta}'_n(\theta_1), \dots, \hat{\beta}'_n(\theta_J))',$$

and let ξ_0 be the corresponding vector of true regression quantiles,

$$\xi_0 \equiv (\beta'_0(\theta_1), \dots, \beta'_0(\theta_J))';$$

for convenience, the $\{\theta_j\}$ are assumed to be unique and in ascending order,

$$0 < \theta_1 < \theta_2 < \dots < \theta_J < 1.$$

The following theorem then gives the asymptotic distribution of $\hat{\xi}_n$ under the previously-stated conditions.

Theorem 2. For the censored regression model (2.1), if Assumptions P.2, R.2, and E.2 hold for each $\theta \in \{\theta_1, \dots, \theta_J\}$, then the vector of censored regression

quantile estimators $\hat{\xi}_n$ is asymptotically normal,

$$M_n^{-1/2} \sqrt{n} (\hat{\xi}_n - \xi_0) \xrightarrow{d} N(0, I),$$

where $M_n^{-1/2}$ is any square root of a matrix M_n which can be written in the partitioned form $M_n \equiv [\omega_{jk} D_m^{-1}]$ for $m \equiv \max\{j, k\}$ and $j, k = 1, \dots, J$, where

$$\omega_{jk} \equiv \left[\frac{\min\{\theta_j, \theta_k\} - \theta_j \theta_k}{f[F^{-1}(\theta_j)] \cdot f[F^{-1}(\theta_k)]} \right],$$

and

$$D_l \equiv \frac{1}{n} \sum_{i=1}^n E\{1(x'_i \beta_0(\theta_l) > 0) \cdot x_i x'_i\}, \quad l = 1, \dots, J.$$

Proof. Again, the proof here is a straightforward modification of the proof of Theorem 1 of Powell (1984). For each θ_j , the vector of left partials of $\sqrt{n} Q_n(\beta; \theta_j)$ is

$$\Psi_n(\beta; \theta_j) \equiv \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_j}(u_i, x_i, \beta), \quad (3.3)$$

where

$$\psi_{\theta_j}(u_i, x_i, \beta) \equiv 1(x'_i \beta > 0) \cdot [\theta_j - 1(u_i - x'_i \delta_j < F^{-1}(\theta_j))] \cdot x_i, \quad (3.4)$$

for $\delta_j \equiv \beta - \beta_0(\theta_j) \equiv (\beta - \beta_0) - F^{-1}(\theta_j) \cdot e^1$, as before. The same argument given in (A.12) through (A.23) of Powell (1984) can be used to establish the 'asymptotic linearity' relation

$$\begin{aligned} \sqrt{n} (\hat{\beta}_n(\theta_j) - \beta_0(\theta_j)) &= \{f[F^{-1}(\theta_j)] \cdot D_j\}^{-1} \cdot \Psi_n(\beta_0(\theta_j); \theta_j) \\ &\quad + o_p(1). \end{aligned} \quad (3.5)$$

Application of the Lindeberg–Feller central limit theorem to (3.5) yields the asymptotic normality of $\hat{\xi}_n$, and the form of its asymptotic covariance matrix follows from the relation

$$E\{[\Psi_n(\beta_0(\theta_j); \theta_j)] \cdot [\Psi_n(\beta_0(\theta_k); \theta_k)]'\} = [\theta_j - \theta_j \theta_k] \cdot D_l, \quad (3.6)$$

for $l \equiv \min\{j, k\}$. ■

Estimation of the asymptotic covariance matrix M_n of the regression quantile estimators $\hat{\xi}_n$ can be based on the same approach given in section 5 of Powell (1984) for the censored LAD estimator. (Weakly) consistent estimates of the matrices $\{D_l\}$ under the conditions of Theorem 2 are given by

$$\hat{D}_l \equiv \frac{1}{n} \sum_{i=1}^n 1(x'_i \hat{\beta}_n(\theta_l) > 0) \cdot x_i x'_i, \quad l = 1, \dots, J, \quad (3.7)$$

while the density function of $\{u_i\}$ at its θ_k th quantile, $f[F^{-1}(\theta_k)]$, can be consistently estimated by

$$\begin{aligned} \hat{f}_n[F^{-1}(\theta_k)] &\equiv \left[\sum_{i=1}^n 1(x'_i \hat{\beta}_n(\theta_k) > 0) \right]^{-1} \\ &\quad \cdot \hat{c}_n^{-1} \sum_{i=1}^n 1(x'_i \hat{\beta}_n(\theta_k) > 0) \cdot 1(0 \leq \hat{u}_i(\theta_k) \leq \hat{c}_n), \end{aligned} \quad (3.8)$$

a 'kernel' density estimator (with uniform kernel) based on the censored regression quantile residuals

$$\hat{u}_i(\theta_k) \equiv y_i - x'_i \hat{\beta}_n(\theta_k). \quad (3.9)$$

The 'window width' sequence $\{\hat{c}_n\}$ is assumed to satisfy

$$\hat{c}_n = o_p(1) \quad \text{and} \quad \hat{c}_n^{-1} = o_p(\sqrt{n}) \quad (3.10)$$

to ensure convergence in probability of $\hat{f}_n(\cdot)$ to $f(\cdot)$. The proof of consistency of $\hat{f}_n[F^{-1}(\frac{1}{2})]$ in Powell (1984) is readily adapted to estimation of the density function at an arbitrary quantile, as considered here.

4. Linear combinations and differences of quantile estimators

Theorem 2 above provides a basis for comparing the relative efficiency of differing censored regression quantile estimators for the slope coefficients of β_0 , and suggests the possibility of finding still more efficient estimators by taking appropriate weighted averages of the $\{\hat{b}_n(\theta_j)\}$. However, these calculations are considerably more complicated for the censored regression model than the corresponding calculation is for the standard linear model. In Koenker and Bassett (1978), the asymptotic covariance matrix of the vector of regression quantiles for the linear model was shown to be of the form $\Omega \otimes \bar{D}$, where $\Omega \equiv [\omega_{jk}]$ for j and $k = 1, \dots, J$ and $\bar{D} \equiv \lim n^{-1} \sum x_i x'_i$; thus their investigation of linear combinations of regression quantiles only depended on

the form of the error density. On the other hand, in the present (censored) case the covariance matrix depends in a complicated way on the distribution of the regressors as well as the unknown parameter vector; the efficiency of a given censored regression quantile estimator $\hat{\beta}_n(\theta)$ involves both the efficiency of estimation of the θ th quantile of u_i (embodied in the ω_{jj} term) and the proportion of data points with $x_i'\beta_0(\theta) > 0$ (reflected in the D_j matrix). Hence the 'best' choice of θ will vary with these other aspects of the model.

Nevertheless, for a given vector $\hat{\xi}_n$ of regression quantile estimators, the optimal estimator of the slope coefficients of β_0 can be computed as a matrix-weighted average of the $\{\hat{\beta}_n(\theta_j)\}$, with weights given by certain combinations of submatrices of the inverse of M_n , the asymptotic covariance matrix of ξ_0 (a consistent estimator of M_n could be used to compute these weights in practice). An interesting open theoretical question is whether a 'fully efficient' estimator of the slope coefficients can be based upon such an average of quantile estimators. Recent papers by Chamberlain (1984) and Cosslett (1984) have investigated the relevant 'information bound' matrix for the censored regression model with unknown error distribution, and the latter paper gives a formula for the bound which involves a continuous combination of matrices of the form given in the expression for D_j , with continuous weights involving the unknown error density. Whether this efficiency bound can be approximated by a matrix-weighted average of regression quantile estimators, computed at a fine grid of quantile points, is an interesting question for further research.

With the censored regression quantile estimators $\{\hat{\beta}_n(\theta_j)\}$ and a consistent estimate of the asymptotic covariance matrix M_n using (3.7) and (3.8) above, large-sample tests of hypotheses concerning the censored regression model (2.1) under the conditions of Theorem 2 and be constructed in the usual manner. In addition the usual normal-theory tests concerning the values of the unknown coefficients β_0 , the conclusion of Theorem 2 permits hypotheses about the stochastic behavior of the error terms $\{u_i\}$ to be checked as well. The null hypothesis that the distribution of the $\{u_i\}$ has a particular parametric form (e.g., that the errors are i.i.d. Gaussian) can be tested by comparing maximum likelihood estimates of the $\{\beta_0(\theta_j)\}$ to the censored regression quantiles, the latter being consistent under both the null and alternative hypotheses but less efficient than the maximum likelihood estimates under the null hypothesis. Details of the computation of such test statistics are given by Hausman (1978) and Ruud (1982).

The censored regression quantiles can also be used to test the assumption of homoskedasticity of the error terms in the manner proposed by Koenker and Bassett (1982) for the standard linear model. In their approach, the null hypothesis that the error terms $\{u_i\}$ are identically distributed is tested against the alternative that the distribution function of each u_i conditional on the regressors x_i varies systematically with x_i , i.e., $\Pr\{u_i < \lambda | x_i\} \equiv F(\lambda | x_i) \neq F(\lambda)$.

More specifically, Koenker and Bassett suggest testing the null hypothesis

$$H_0: F^{-1}(\theta_j|x_i) = F^{-1}(\theta_j), \quad j = 1, \dots, J, \quad (4.1)$$

against a sequence of linear alternatives

$$H_A: F^{-1}(\theta_j|x_i) \equiv (1 + x_i'\gamma_n) \cdot F^{-1}(\theta_j), \quad j = 1, \dots, J, \quad (4.2)$$

where $\{\gamma_n\}$ is some sequence of conformable vectors and $F^{-1}(\theta)$ is a proper inverse cumulative function.¹ The null and alternative hypotheses can be rewritten in terms of the $\{\beta_0(\theta_j)\}$:

$$H_0: B_0(\theta_j) = \beta_0 + F^{-1}(\theta_j) \cdot e^1, \quad j = 1, \dots, J, \quad (4.1')$$

versus

$$H_A: \beta_0(\theta_j) = \beta_0 + F^{-1}(\theta_j) \cdot [e^1 + \gamma_n], \quad j = 1, \dots, J. \quad (4.2')$$

The test statistic proposed by Koenker and Bassett (1982) is actually a test of a particular implication of H_0 , namely, the homogeneous linear hypothesis

$$H_0^*: H\xi_0 = 0, \quad (4.3)$$

where premultiplication of the (quantile) vector ξ_0 by the matrix H yields a vector of slope coefficients of the differences $[\beta_0(\theta_j) - \beta_0(\theta_{j-1})]$, $j = 1, \dots, J$.² With a consistent estimate of the asymptotic covariance matrix of $\hat{\xi}_n$, as well as the algebraic form of the matrix H [given in section 4.1 of Koenker and Bassett (1982)], a Wald-test statistic for H_0^* which has a limiting chi-square distribution under the null hypothesis can be constructed.

Koenker and Bassett point out, however, that the form of the asymptotic covariance matrix of the vector of (censored) regression quantile estimators under the alternative hypothesis will depend in general on the specific sequence $\{\gamma_n\}$ which defines the alternative H_A . If the sequence of alternatives is $O(n^{-1/2})$ – say, $\gamma_n = n^{-1/2}\gamma_0$, as Koenker and Bassett assume – then the result of Theorem 2 will still hold under the alternative hypothesis, provided

¹ It is implicitly assumed that $\{\gamma_n\}$ and the $\{x_i\}$ are such that $x_i'\gamma_n > -1$ for all i and n , so that the monotonicity of the conditional quantiles is preserved.

² This is weaker than H_0 of (4.1) in that the order relationships for the intercept terms are not imposed.

ξ_0 is redefined to be

$$\begin{aligned}\xi_0 &= [\beta'_0 + F^{-1}(\theta_1) \cdot [e^1 + \gamma_n]', \dots, \beta'_0 + F^{-1}(\theta_J) \cdot [e^1 + \gamma_n]']' \\ &\equiv \xi_0(n).\end{aligned}\quad (4.4)$$

On the other hand, if the alternatives do not vary with n , so that $\gamma_n = \gamma_0$ for all n , then the expression for the asymptotic covariance matrix of $\hat{\xi}_n$ given in the statement of Theorem 2 will be incorrect, although asymptotic normality of $\hat{\xi}_n$ can still be shown to hold. More generally, suppose the conditional θ th quantile of u_i given x_i has the (fixed) linear form

$$F^{-1}(\theta|x_i) \equiv x_i' \gamma(\theta); \quad (4.5)$$

then the θ th conditional quantile of the censored dependent variable will still be $\max\{0, x_i' \beta_0(\theta)\}$, where now

$$\beta_0(\theta) \equiv \beta_0 + \gamma(\theta). \quad (4.6)$$

[Definition (2.4) above is clearly a special case, with $\gamma(\theta) = F^{-1}(\theta)e^1$.] If Assumptions P.2, R.2, and E.2 are weakened to satisfy (4.5) and (4.6), then the result of Theorem 1 still holds, i.e., $\hat{\beta}_n(\theta) \rightarrow \beta_0(\theta)$ a.s., and a modification of the proof of Theorem 2 yields

$$[M_n^*]^{-1/2} \sqrt{n} (\hat{\xi}_n - \xi_0) \xrightarrow{d} N(0, I), \quad (4.7)$$

where the matrix M_n^* can be expressed in the partitioned form

$$M_n^* \equiv [(\theta_m - \theta_j \theta_k) \cdot G_j^{-1} D_m G_k^{-1}], \quad m \equiv \min\{j, k\}, \quad (4.8)$$

where D_j is defined in the statement of Theorem 2 and

$$G_j \equiv \frac{1}{n} \sum_{i=1}^n E_x \{ f_i [x_i' \gamma(\theta)] \cdot 1(x_i' \beta_0(\theta) > 0) \cdot x_i x_i' \}, \quad (4.9)$$

for $f_i(\cdot)$ the conditional density of u_i given x_i . The matrices $\{G_j\}$ can be consistently estimated by

$$\hat{G}_j = (\hat{c}_n \cdot n)^{-1} \sum_{i=1}^n 1(x_i' \hat{\beta}_n(\theta_j) > 0) \cdot 1(0 \leq \hat{u}_i(\theta_j) \leq \hat{c}_n) \cdot x_i x_i', \quad (4.10)$$

where $\hat{u}_i(\theta_j)$ and $\hat{c}_n$ satisfy conditions (3.9) and (3.10) above. If heteroskedas-

ticity of the form (4.5) above is suspected, estimates of the asymptotic covariance matrix of $\hat{\xi}_n$ based on (4.8) would be preferred, since this expression is valid in either the homoskedastic or heteroskedastic cases.

Conditional symmetry of the error distribution about zero can be tested in the same way. In the context of the linear conditional quantiles of (4.5) and (4.6), the hypothesis of conditional symmetry can be expressed as

$$H_0: \gamma(\theta) = -\gamma(1 - \theta), \quad (4.11)$$

which implies $\gamma(\frac{1}{2}) = 0$. This null hypothesis implies linear restrictions of the form (4.3), where now premultiplication of ξ_0 by H yields 'symmetric second differences' of the form $[\hat{\beta}_n(\theta) - \hat{\beta}_n(\frac{1}{2})] - [\hat{\beta}_n(\frac{1}{2}) - \hat{\beta}_n(1 - \theta)]$ for $\theta > \frac{1}{2}$. If the error terms u_i are presumed to be heteroskedastic, the hypothesis of conditional symmetry is crucial to the interpretation of β_0 as the coefficients of the 'typical' response of the censored dependent variable to the regressors, so failure of both heteroskedasticity and conditional symmetry tests would suggest a substantial respecification of model (2.1).

References

- Amemiya, T., 1982, Two stage least absolute deviations estimators, *Econometrica* 50, 689–711.
- Bloomfield, P. and W.L. Steiger, 1983, *Least absolute deviations: Theory, applications, and algorithms* (Birkhauser, Boston, MA).
- Chamberlain, G., 1984, Asymptotic efficiency in semiparametric models with censoring, Manuscript, May (Department of Economics, University of Wisconsin, Madison, WI).
- Cosslett, S.R., 1984b, Efficiency bounds for distribution-free estimators of the binary choice and the censored regression models, Manuscript, June (Center for Econometrics and Decision Sciences, University of Florida, Gainesville, FL).
- Hausman, J.A., 1978, Specification tests in econometrics, *Econometrica* 46, 1251–1271.
- Koenker, R. and G.S. Bassett, Jr., 1978, Regression quantiles, *Econometrica* 46, 33–50.
- Koenker, R. and G.S. Bassett, Jr., 1982, Robust tests for heteroscedasticity based on regression quantiles, *Econometrica* 50, 43–61.
- Manski, C.F., 1975, Maximum score estimation of the stochastic utility model of choice, *Journal of Econometrics* 3, 205–228.
- Manski, C.F., 1984, Semiparametric analysis of discrete response: Asymptotic properties of the maximum score estimator, SSRJ workshop paper no. 8413 (University of Wisconsin, Madison, WI).
- Obenhofer, W., 1982, The consistency of nonlinear regression minimizing the L_1 norm, *Annals of Statistics* 10, 316–319.
- Powell, J.L., 1983, The asymptotic normality of two-stage least absolute deviations estimators, *Econometrica* 51, 1569–1575.
- Powell, J.L., 1984, Least absolute deviations estimation for the censored regression model, *Journal of Econometrics* 25, 303–325.
- Ruud, P., 1982, A score test of consistency, Manuscript, Feb. (University of California, Berkeley, CA).
- Tobin, J., 1958, Estimation of relationships for limited dependent variables, *Econometrica* 26, 24–36.